

OAD: An Unsupervised 3D Masked-Autoencoder for Coastal Ocean Anomaly Detection, Validated Against In-Situ ESP Biology

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Abstract

Harmful algal blooms along the U.S. Pacific Northwest coast are visible in satellite imagery, but using single satellite channels like chlorophyll-a to predict downstream biology has historically not worked very well. We train a 3D convolutional masked-autoencoder on 22 years (2003–2024) of MODIS Aqua imagery over the 41–49° N coastal box. The model takes four ocean channels (chlorophyll-a, water clarity, fluorescence, sea-surface temperature) on a 321×409 grid and a 4-day time window, and outputs one number per day per region: an “anomaly score” based on how badly it reconstructs the input. We do not give it any bloom labels.

We then check whether that score is useful. Three results: (1) at a 7-day forecast horizon it is the only method we tested with positive R^2 in every region (0.10–0.26)—a modest absolute number, but the matched-complexity PCA baseline sits at zero. (2) The score correlates with in-water Environmental Sample Processor measurements at the NEMO mooring during 2016–2018 (Pseudo-nitzschia cells $r=+0.46$, $p=3\times 10^{-5}$; particulate domoic acid $r=+0.33$, $p=1\times 10^{-3}$). These are not strong correlations, but they are statistically significant for an unsupervised signal validated against an instrument it never trained on. (3) The summer-mean score in 2014 is the highest of any year on record in three of four regions and the second-highest in the fourth, matching the documented Pacific marine heatwave. We also document a moderate cloud-cover confound (19–24% of score variance) and suggest that downstream users carry the in-region valid-pixel fraction as a parallel feature.

1. Introduction

Toxin-producing Pseudo-nitzschia blooms happen every year along the U.S. Pacific Northwest (PNW) coast. The toxin they make, domoic acid, builds up in razor clams and shuts down beach harvests [10, 16]. These blooms are not invisible from space—MODIS Aqua sees the elevated chlorophyll, the warmer-than-average water, the brighter fluorescence, the changes in water clarity. The problem is that picking out which patches of green water will actually produce toxin from those satellite channels has historically been hard [1, 6]. Blooms look different from each other, only some species produce toxin, and PNW cloud cover blocks the sensor a lot of the time.

We try a different angle. Instead of training a model to predict a specific target (which needs labels, and bloom labels are expensive), we ask whether an unsupervised model can learn a useful summary of “what the ocean looks like” from the satellite record alone. Masked autoencoders [4] are a self-supervised approach where the model has to fill in chunks of the image that have been hidden; in natural images this forces it to learn higher-level structure than just local texture. We adapt that idea to coastal satellite imagery, where the signal is biophysical (chlorophyll, temperature) rather than categorical (cat vs. dog).

We call the resulting system OAD (Ocean Anomaly Detection). It is a 3D convolutional masked-autoencoder trained on 22 years of MODIS Aqua over the PNW coast (Fig. 1). The output is a single number per region per day, intended as a starting point for downstream HAB tools. Our contributions:

1. A 3D convolutional autoencoder with three-phase masked-autoencoder training that turns the MODIS cube into one daily score per coastal region.
2. A lead-time comparison showing OAD is the only

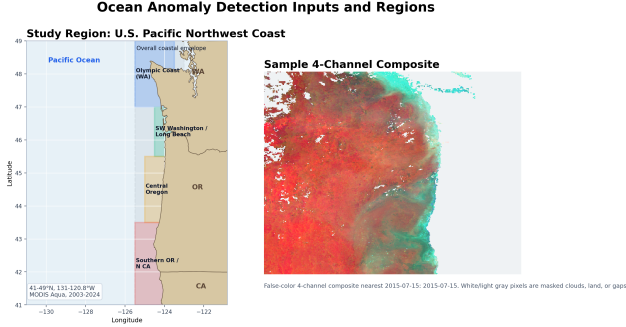


Figure 1. Study region (41–49°N) with example MODIS Aqua 8-day composite. The U.S. PNW coastal box is analyzed at 0.025° resolution on a 321×409 grid. Five alongshore sub-regions are used for downstream analysis; the Olympic Coast region contains the NEMO mooring (47.97°N, 124.97°W) at which ESP in-situ measurements were collected during the 2016–2018 ChaBa deployments.

method tested that is better than predicting the mean at a 7-day horizon in every region, while two natural baselines (matched-complexity PCA and day-of-year climatology) are not.

3. Comparison against in-water Environmental Sample Processor data at the NEMO mooring (2016–2018): the unsupervised satellite score correlates with both Pn cell density ($r=+0.46$) and particulate domoic acid ($r=+0.33$), with bootstrap 95% confidence intervals that exclude zero.
4. The summer-mean score for 2014 is the highest of any year in the record in three out of four regions, lining up with the documented Pacific marine heat-wave, again without ever being told which year that was.

2. Related Work

Satellite remote sensing for HABs. Operational monitoring has long used satellite chlorophyll-a as a near-real-time bloom indicator [14, 19], but in PNW coastal water the relationship between chl-a and beach-level toxin outcomes is weak [15, 16]. Multi-channel approaches that combine SST, fluorescence, and clarity perform incrementally better [1], but the per-pixel single-day framing remains limiting.

Self-supervised learning for Earth observation. Masked autoencoders [4] train a network to reconstruct large hidden fractions of the input, forcing learning of higher-level structure. In Earth observation, SatMAE [3] and follow-up work [17] apply the principle to optical satellite imagery, primarily for classification. We adapt it to a 3D (time-augmented) coastal ocean cube where the

temporal axis lets the model learn dynamics alongside spatial structure—a regime where the dominant signal is biophysical, not land-cover categorical.

In-situ validation via ESP/ChaBa. The Environmental Sample Processor [13] is an autonomous in-water sampler deployed at the NEMO mooring during ChaBa [11], measuring Pn cell density (sandwich hybridization) and particulate domoic acid (competitive ELISA) at daily cadence. These datasets are our independent biological validation target.

3. Method

Data. The training cube is built from MODIS Aqua Level-3 daily ocean-color products [12] covering 2003–2024 (~4,700 daily rolling 8-day composites) on a 321×409 grid at 0.025° resolution. Four channels are used: chl-a, Kd490 (water clarity), normalized fluorescence (nFLH), and SST. A 2D static ocean/land mask excludes land pixels; NaN pixels (cloud-flagged or otherwise unobserved) are excluded per-pixel from the masked-MSE reconstruction loss.

Architecture. OAD is a 3D ConvAE operating on tensors of shape ($C=4$, $T=4$, $H=321$, $W=409$) (Fig. 2). The encoder applies four blocks of 3D convolution with stride-2 spatial downsampling, GroupNorm, and GELU activation, producing a latent of dimension 32. The decoder mirrors the encoder with transposed convolutions and upsampling.

Three-phase training. Phase A (plain autoencoder) produced reconstruction error that correlated poorly with bloom events—the model learned to copy textures. Phase B added per-region z-score post-processing, which improved scale but not the underlying representation. Phase C is masked-autoencoder pretraining [4] with 50% random pixel masking, forcing the model to infer hidden pixels from larger spatial context. We select Phase C with 50% masking (mae050) over a 70%-masking variant because mae050 has lower cloud confound (Sec. 4.4) without sacrificing multi-step forecast skill.

Score derivation. For each daily composite at region R , the trained model produces a per-pixel reconstruction. We average per-pixel reconstruction error over in-region ocean pixels and z-score the result against an in-region historical distribution, producing a single scalar per (region, day). Downstream analyses operate only on this per-region daily scalar; the convolutional model is not invoked again at consumption time.

4. Results

The OAD score is evaluated along three independent axes: intrinsic lead-time forecastability against

3D Masked Autoencoder Anomaly-Detection Pipeline

Train on masked MODIS structure, then aggregate reconstruction error into regional anomaly scores.

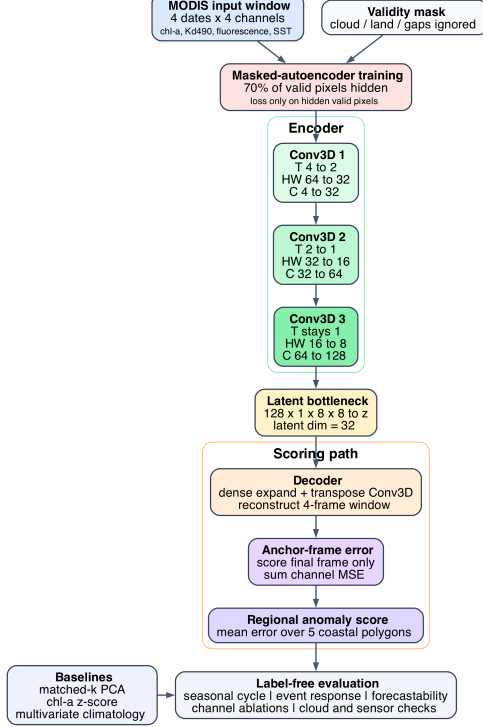


Figure 2. OAD 3D convolutional masked-autoencoder. Input is a 4-channel ($C=4$), 4-day-window ($T=4$), 321×409 tensor. The encoder applies four stride-2 3D conv blocks to a latent of dimension 32; the decoder symmetrically reconstructs the input. During Phase C training, 50% of input pixels are randomly masked and the reconstruction loss is computed only on ocean pixels with cloud-flagged NaN pixels excluded per-pixel.

baselines (Sec. 4.1), independent in-situ biological validation at the NEMO mooring (Sec. 4.2), and consistency with documented PNW climate events (Sec. 4.3). A short cloud-confound characterization follows (Sec. 4.4).

4.1. Lead-time forecastability vs. baselines

We compare OAD against a matched-dimensionality PCA baseline and a day-of-year climatology baseline, forecasting per-region anomaly state from the past four days on the 2019+ held-out period. Tab. 1 reports results for SW Washington (the highest-amplitude region); Fig. 3 shows the cross-region picture at the 7-day horizon.

At the honest 7-day lead, OAD is the only method we tested with positive R^2 in every one of the five PNW regions (range 0.10–0.26). PCA, with the same number of dimensions, drops to roughly zero in every re-

Table 1. Lead-time forecastability R^2 , SW Washington region (2019+ held-out). The 1-day result is inflated by 8-day composite overlap (consecutive composites share 7/8 input days); the honest comparison is at lead ≥ 7 days.

Lead	OAD-50	OAD-70	PCA	Climatology
1 day	+0.84	+0.87	−0.11	≤ -0.5
7 days	+0.15	+0.15	−0.11	< 0
14 days	+0.05	+0.05	−0.11	< 0

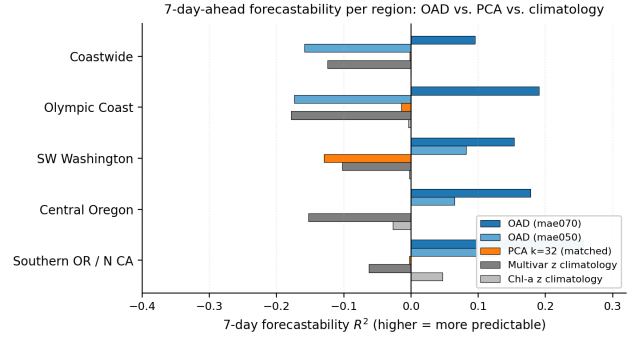


Figure 3. Can each method predict the next week’s regional anomaly state from the past 4 days? Bars are held-out R^2 on data from 2019 onward. Bars to the right of zero do better than just guessing the mean; bars to the left do worse. OAD-70 (dark blue) is positive in all five regions (0.10–0.26)—not a huge number, but every other method we tried sits at or below zero. PCA (orange) is the right control because it has the same number of dimensions as the autoencoder, so the gap is not just “more parameters help.” The lower-masking OAD-50 variant we ship downstream is positive in 3 of 5 regions; it trades a little R^2 for a cleaner cloud-cover signature (Sec. 4.4).

gion; the day-of-year climatology baseline actively does worse than predicting the mean in most regions (Fig. 3). To be clear: an R^2 of 0.10–0.26 is not great in absolute terms—most of the variance in the next week’s ocean state remains unexplained. The point is the gap between OAD and the matched-complexity baselines, not the absolute magnitude. That gap is the evidence that the autoencoder is picking up something beyond what a linear projection of the same pixels can find.

4.2. In-situ validation against ESP/ChaBa

The cleanest test of whether OAD captures real biology—rather than an idiosyncratic artifact of the unsupervised training objective—is to compare it against independent in-situ measurements at the same time and place. We use ESP-derived Pn cell density and particulate domoic acid (pDA) from the 2016–2018 ChaBa deployments at the NEMO mooring (47.97°N , 124.97°W) [11]. Pearson correlations with 95% non-

Table 2. OAD score vs. in-situ ESP measurements at NEMO, 2016–2018. Pn cells: $N=76$ days (SHA assay). pDA: $N=90$ days (cELISA assay). The strongest Pn-cell correlation appears in the region containing NEMO directly (Olympic Coast); the strongest pDA correlation appears in the downwind transport corridor (SW Washington).

Region	Pn cells r [CI]	pDA r [CI]
Olympic Coast	+0.46 [.19, .66]	+0.32 [−.02, .53]
SW WA	+0.31 [.11, .51]	+0.33 [.13, .52]
Coastwide envelope	+0.16 [−.10, .39]	+0.21 [.02, .36]

parametric bootstrap CIs (2,000 resamples, seed = 42) appear in Tab. 2.

The score correlates with both biological measurements, and the correlations are statistically significant: the strongest Pn-cell correlation is in Olympic Coast (which contains NEMO; $p = 3 \times 10^{-5}$), and the strongest pDA correlation is in SW Washington just downwind ($p = 1 \times 10^{-3}$). Bootstrap 95% confidence intervals exclude zero in both cases. Honest framing: $r \approx 0.33$ –0.46 is moderate, not strong—there is plenty of bloom-day-to-bloom-day biology that the satellite is missing. But the model was never trained on these in-water measurements, so any consistent signal is informative.

The coastwide envelope is the weakest of the three regions tested for both targets, which is what we would expect: spatial averaging over a 41–49°N strip dilutes the local Olympic Coast bloom. The per-day scatter (Fig. 4) shows the correlation visually, and the time-series overlay (Fig. 5) shows that OAD peaks and ESP peaks tend to move together during each of the five 2016–2018 deployment windows. The 2017 spring deployment is the clearest case: OAD ramps up several days before the ESP-measured Pn peak, which suggests it may be useful as a leading indicator for when to deploy in-water samplers.

4.3. Agreement with documented PNW climatology

The unsupervised score also recovers well-established PNW coastal ocean patterns without ever having been told about specific events. Three patterns emerge.

Seasonal cycle. The score rises sharply in late spring (April–May), peaks in mid-summer, and declines through autumn across all five regions—matching canonical PNW Pn bloom phenology [8, 15, 16]. Amplitude is largest in SW Washington (Columbia River plume $\times$ shelf upwelling) and Olympic Coast (Juan de Fuca eddy retains nutrient-rich water [5]).

The 2014 marine heatwave stands out. The clearest test of the score’s climate sensitivity is whether it picks

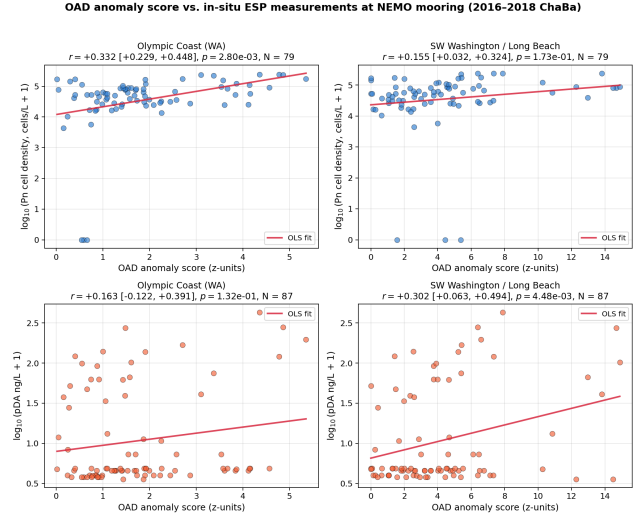


Figure 4. Day-by-day scatter of the unsupervised OAD score against the in-water biology measured at the NEMO mooring. Top row: Pn cell density measured by ESP. Bottom row: particulate domoic acid measured by ESP. Each dot is one matched day during the 2016–2018 ChaBa deployments; red line is a linear fit for visualization. Pearson r and bootstrap 95% CIs are computed on raw values; the axes are log-scaled only for readability. The Pn-cell signal is strongest at the region that contains NEMO (Olympic Coast, top-left), while the toxin signal is strongest in the SW Washington region just downwind—which is the direction the bloom typically gets carried along the coast.

out the year of the 2014–16 Pacific marine heatwave (“the Blob” [2]), which triggered the largest *P. australis* bloom on record [9, 10]. Fig. 6 plots the May–September mean OAD score for each year. In Olympic Coast, Central Oregon, and Southern OR/N. California, 2014 is the highest summer-mean of all 19 measured years; in SW Washington it is the second-highest. The model was never told which years had blooms, so this ranking comes entirely from the satellite imagery itself.

Cross-region coherence with heterogeneity. Regional scores correlate at > 0.5 on monthly means—consistent with PDO-modulated basin-scale upwelling and ENSO teleconnections [7]—but each region preserves its own high-frequency signal from local eddies, plumes, and wind events. This dichotomy is exactly the structure a representation faithful to PNW coastal dynamics should produce.

4.4. Cloud-cover confound

Per-region Pearson correlation between the OAD score and the in-region valid-pixel fraction is moderate: SW Washington shows $r \approx +0.44$ ($\sim 19\%$ of variance

OAD score (blue line) vs. ESP in-situ measurements (markers) at NEMO during 2016–2018 ChaBa deployments

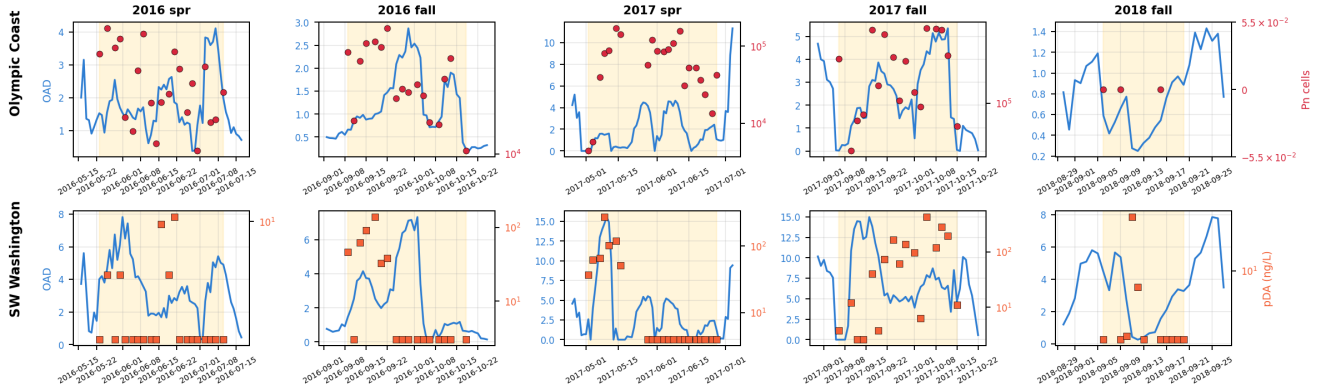


Figure 5. Day-by-day overlay of the OAD satellite score (blue line, left axis) and the in-water ESP measurements (red/orange markers, right axis) during the five 2016–2018 ChaBa deployments at NEMO. Yellow bands mark each deployment window. Top row: Olympic Coast OAD vs. ESP Pn cells per liter (log scale). Bottom row: SW Washington OAD vs. ESP particulate domoic acid in ng/L (log scale). When the satellite score goes up, the in-water biology tends to go up too. The 2017 spring panel (third column) is the cleanest example: OAD starts rising several days before the ESP measures the Pn peak, which is the kind of behavior you would want from an early-warning signal.

Summer (May–Sep) mean OAD score per region, 2003–2024

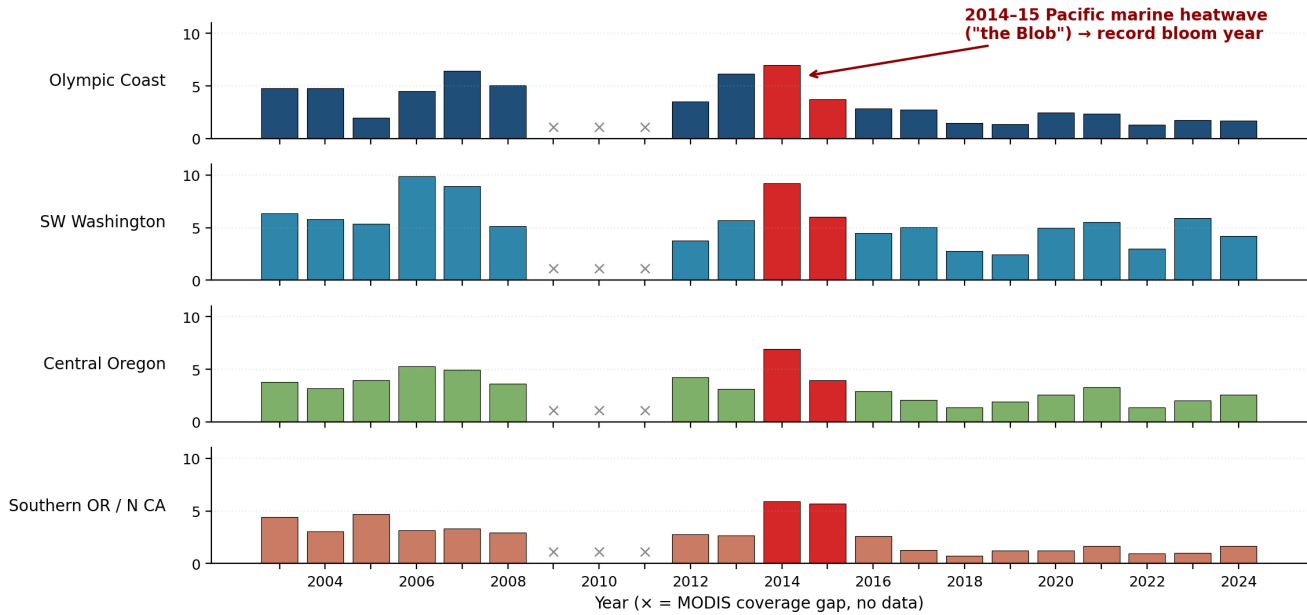


Figure 6. Summer (May–September) mean OAD score for each year, by region. Red bars mark 2014–15, the peak years of the Pacific marine heatwave (“the Blob”). 2014 is the highest summer mean of any year on record in three of the four regions, and the second-highest in the fourth—without the model ever being shown a bloom-year label. The “x” marks the 2009–11 MODIS Aqua coverage gap (no data).

covaries with cloud cover) and Olympic Coast shows $r \approx +0.49$ ($\sim 24\%$). The mae070 variant is 4–10 percentage points higher in absolute value, which is why we selected mae050 as the reporting checkpoint. Down-

stream applications should carry the in-region valid-pixel fraction as a parallel feature so consuming models can learn to discount cloudy weeks.

5. Discussion

The three checks each show something modest but real. The forecastability R^2 is small in absolute terms (0.10–0.26), but every baseline we tried at the same complexity drops to zero or below—so the gap, not the absolute number, is the result. The ESP correlations ($r = 0.33$ – 0.46) are moderate and significant, but the model was never trained on those measurements, so any consistent signal is informative. The 2014 heatwave-year ranking comes out cleanly without any temporal labels. None of this proves the score is a finished bloom-prediction tool. It does show that an unsupervised representation of the satellite cube is picking up structure that linear baselines miss and that lines up with real biology at the source.

What this paper does not show is whether the source-region OAD score predicts beach-level shellfish toxicity. That depends on a chain of additional physics: wind direction, onshore transport, which Pn species is present and whether it is producing toxin, and how much each clam happens to filter. We test only the offshore signal here.

Limitations. (1) MODIS Aqua sensor sunset will require recalibration to PACE [18]; (2) the 2009–2011 MODIS coverage gap leaves wide uncertainty in some regions; (3) the cloud-confound is partial (~ 19 – 24% variance); (4) in-situ validation uses only one mooring (NEMO) with 76–90 daily samples; (5) regions are hand-defined—learning regions via clustering on the latent is a natural extension.

6. Conclusion

We trained an unsupervised 3D convolutional masked-autoencoder on 22 years of MODIS Aqua imagery over the U.S. Pacific Northwest coast, and we use it to produce one number per region per day. That number does three useful things: it forecasts the next week’s regional ocean state better than two natural baselines (modest R^2 , but the baselines are at zero); it correlates with in-water ESP measurements of Pseudo-nitzschia cells and particulate domoic acid at the NEMO mooring, even though the model never saw those measurements during training; and its summer mean for 2014 is the highest of any year on record in 3 of 4 regions, matching the documented Pacific marine heatwave. The results are not blockbusters, but every line of evidence is consistent with the score carrying real ocean information that simpler methods miss. Code and trained checkpoints are at <https://github.com/ansoncchen/DATect-Forecasting-Domoic-Acid> under ocean anomaly detection/.

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